EXPERIMENT - 5

Aim: To create a Class Diagram for <case study name>

Prerequisite:

1. Basic understanding of functionality for the system to be designed

Outcome: After successful completion of this experiment students will be able to

1. Identify the various Classes and their association with each other.
2. Identify the various attributes and functionality of individual classes.

Theory:

What is Data modeling?

Data modeling in software engineering is the process of creating a data model for an information system by applying formal data modeling techniques.

Data modeling is a process used to define and analyze data requirements needed to support the business processes within the scope of corresponding information systems in organizations. Therefore, the process of data modeling involves professional data modelers working closely with business stakeholders, as well as potential users of the information system.

There are three different types of data models produced while progressing from requirements to the actual database to be used for the information system.[2] The data requirements are initially recorded as a conceptual data model which is essentially a set of technology independent specifications about the data and is used to discuss initial requirements with the business stakeholders. The conceptual model is then translated into a logical data model, which documents structures of the data that can be implemented in databases. Implementation of one conceptual data model may require multiple logical data models. The last step in data modeling is transforming the logical data model to a physical data model that organizes the data into tables, and accounts for access, performance and storage details. Data modeling defines not just data elements, but also their structures and the relationships between them.[3]

Data modeling techniques and methodologies are used to model data in a standard, consistent, predictable manner in order to manage it as a resource. The use of data modeling standards is strongly recommended for all projects requiring a standard means of defining and analyzing data within an organization, e.g., using data modeling:

- to assist business analysts, programmers, testers, manual writers, IT package selectors, engineers, managers, related organizations and clients to understand and use an agreed semi-formal model the concepts of the organization and how they relate to one another
- to manage data as a resource
- for the integration of information systems
- for designing databases/data warehouses (aka data repositories)

Data modeling may be performed during various types of projects and in multiple phases of projects. Data models are progressive; there is no such thing as the final data model for a business or application. Instead a data model should be considered a living document that will change in response to a changing business. The data models should ideally be stored in a repository so that they can be retrieved, expanded, and edited over time. Whitten (2004) determined two types of data modeling:

- Strategic data modeling: This is part of the creation of an information systems strategy, which defines an overall vision and architecture for information systems is defined. Information engineering is a methodology that embraces this approach.
- Data modeling during systems analysis: In systems analysis logical data models are created as part of the development of new databases.

Data modeling is also used as a technique for detailing business requirements for specific databases. It is sometimes called database modeling because a data model is eventually implemented in a database.

1) How to identify the Classes for a scenario.

   For a scenario, Classes can be identified using these six class selection characteristics

   a. Retained information

      i. Information must be remembered about the system over time

   b. Needed services

i. Set of operations that can change the attributes of a class

c. Multiple attributes
    i. Whereas, a single attribute may denote an atomic variable rather than a class

d. Common attributes
    i. A set of attributes apply to all instances of a class

e. Common operations
    i. A set of operations apply to all instances of a class

f. Essential requirements
    i. Entities that produce or consume information

2) Guidelines for selecting appropriate classes and deciding their attributes and functions.

Attributes and operations can be selected using the following steps:

- Attributes of a class are those nouns from the grammatical parse that reasonably belong to a class
- Attributes hold the values that describe the current properties or state of a class
- An attribute may also appear initially as a potential class that is later rejected because of the class selection criteria
- In identifying attributes, the following question should be answered
    - What data items (composite and/or elementary) will fully define a specific class in the context of the problem at hand?
- Usually an item is not an attribute if more than one of them is to be associated with a class
    - Four categories of operations
    - Operations that manipulate data in some way to change the state of an object (e.g., add, delete, modify)
    - Operations that perform a computation
    - Operations that inquire about the state of an object
    - Operations that monitor an object for the occurrence of a controlling event

- An operation has knowledge about the state of a class and the nature of its associations
- The action performed by an operation is based on the current values of the attributes of a class
    - Using a grammatical parse again, circle the verbs; then select the verbs that relate to the problem domain classes that were previously identified

PART B-

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Procedure:

1) List down the various classes for the system according to the guidelines provided.
2) Decide the attributes for these Classes and the relationships between them.
3) Prepare the Class diagram.

Based on the detailed class diagram we just built for the Patient Care platform, here is the complete list of the classes categorized by their role in the system:

Abstract Base Classes

- **User:** The foundational class for all individuals accessing the system.
- **HealthRecord:** The foundational class for any logged medical data.

User Role Classes (Inherit from User)

- **Patient:** Represents the primary user managing their health.
- **Doctor / Dietician:** Represents the clinical professionals treating the patient.
- **Caretaker / Nurse:** Represents the personnel managing daily care and medications.
- **Family (Patient Family):** Represents the relatives overseeing the caregiver logs.

Health Tracking Classes (Inherit from HealthRecord)

- **VitalsLog:** Handles standard daily health metrics (e.g., blood pressure, temperature).
- **MenstrualLog:** Handles reproductive health and cycle tracking.

Core System Entities

- **Appointment:** Manages scheduling and booking between patients and doctors.
- **Medication:** Manages prescription details, dosages, and reminder schedules.

Class Diagram:

